

Note-Progress in edge plasma turbulence modelling hierarchy of models from 2D transport application to 3D fluid simulations in realistic tokamak geometry

Information

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Abstract

This contribution presents the recent effort at CEA and French federation for Fusion to simulate edge plasma transport with the new code SOLEDGE3X. The latter can be used both as a 2D transport code or as a 3D turbulence code. It makes possible simulating edge plasma up to the first wall including the complex wall geometry. It also includes neutral recycling and impurity sputtering, seeding and transport. In order to improve turbulence description in transport simulation, a reduced model for turbulence intensity prediction has been derived and implemented, based on “k-epsilon” like models from the neutral fluid community. Applications to a JET L-mode detached plasma and to a WEST plasma are used as illustration of the code abilities.

Idea

The authors of this article intend to use the reduced model (k-epsilon model) to compute the 2D/3D plasma turbulence, because the main drawback of the classical computational methods with ad-hoc diffusion is lack of predictability due to the loss of some basic turbulence mechanism.

What do I learn/understand ?

The k-epsilon model is a *heuristic* turbulence model and it is used to obtain some information about turbulence in the mean field transport simulation. This model also relies on other equations, which depends on the objective.

The classical mean field approach has some drawbacks: the experimental measurements contra-

dict the local approximation; the level of fluctuations in the plasma is required.

There are some important observations of the simulation with reduced model:

- “Figure 3 shows a comparison between simulation results and experimental measurements. The overall gradients are well recovered by the k-epsilon model. Figure 4 shows quantities computed by the reduced turbulence model. One recovers a ballooning of the turbulence intensity on the low field side. Turbulence propagates on the high field side by parallel transport.”
- “Figure 5 compares radiation location between simulation and experimental tomographic reconstruction. The radiation seems more concentrated around the Xpoint in the simulation compared to the experimental radiation map. This may be due to the absence of radiating impurities in the simulation. Also, the density peaking in the closed flux surface region is not caught by the simulation. A radial pinch might be needed to recover this feature.”

There are some related and interesting topics, such as the discussion of validity of local approximation, and supplementary equations with k-epsilon model to obtain more information about plasma turbulence.

Reference

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